Fast Differential Mutual Information Inference with a Bias-Corrected Welch–Satterthwaite Reference

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Abstract

Mutual information (MI) quantifies arbitrary statistical dependence, yet standard significance procedures are primarily designed to test independence within one population. A different scientific question arises when two independent populations may have distinct joint distributions but the same strength of dependence: $H_0 : \mathrm{I}(P) = \mathrm{I}(Q)$, allowing $P \neq Q$. Resampling can address this weak null but is computationally costly, whereas a bias-corrected influence-function Wald test is fast but can be liberal in finite samples. We propose a deterministic refinement, termed *Daniel's Method*, that combines the classical first-order correction for discrete plug-in MI, an empirical influence-function standard error, and a Welch–Satterthwaite-inspired Student reference. The estimator, standard error, and test statistic are unchanged from the normal Wald baseline; only the reference distribution and confidence-interval critical value change.

The method was evaluated on 1.22 million weak-null table pairs and 50,000 power pairs spanning square and rectangular alphabets from 2×2 to 20×20, balanced and skewed margins, and sample-size ratios up to 1:4. At nominal $\alpha = 0.05$, mean absolute false-positive-rate error fell from 0.01177 to 0.01084 in the targeted hard grid, a 7.9% relative improvement, while broad-grid error changed from 0.00514 to 0.00504. An independently generated holdout confirmed the direction of the effect. Mean power loss was 0.00154, and median runtime was 0.128 ms versus 0.117 ms for the normal test. Studentized permutation remained better calibrated (hard-grid error 0.00642) but was substantially slower. An adversarial audit showed that the implemented component degrees of freedom are heuristic for plug-in influence variances, so the procedure is not an exact finite-sample Welch test. The evidence supports a low-cost, mildly conservative refinement for regular differential-MI inference, not a universal solution to sparse tables or independence.

Keywords: mutual information; differential mutual information; Welch–Satterthwaite approximation; influence function; finite-sample inference; permutation test

1 Introduction

Mutual information (MI) is a model-free measure of statistical dependence introduced by Shannon [1]. For categorical variables it summarizes the departure of a joint distribution from the product of its margins and detects associations that need not be linear, monotone, or low-dimensional. These properties make MI attractive in neuroscience, genomics, ecology, remote sensing, and complex-systems research [2, 3].

Most familiar MI significance procedures answer a one-population question: are X and Y independent, or equivalently, is $\mathrm{I}(X;Y) = 0$? The likelihood-ratio statistic $2n\widehat{\mathrm{I}}$ has an asymptotic chi-squared reference for fixed categorical alphabets, and permutation procedures commonly shuffle one variable to destroy dependence. Toolkits such as JIDT make these calculations accessible [3].

Many comparative studies require a different test. Suppose the same categorical variables are measured in two independent populations, such as treatment and control, two environments, or two time periods. The question is then

$$H_0 : \mathrm{I}(P) = \mathrm{I}(Q) \quad \text{against} \quad H_1 : \mathrm{I}(P) \neq \mathrm{I}(Q), \qquad (1)$$

where the full joint distributions P and Q may differ. This is a *weak null*: equal MI does not imply equal margins, cell probabilities, or association patterns. A chi-squared independence test in either table does not answer Eq. (1). Likewise, JIDT's standard shuffled independence significance API is not a direct comparator.

The usual analytic route treats MI as a smooth multinomial functional away from independence. A delta-method or influence-function calculation gives an asymptotically normal estimator, and two independent variance contributions can be added. For finite samples, however, the plug-in MI estimate is biased upward and its estimated uncertainty is itself noisy. Unequal sample sizes make the leading bias differ between groups. A normal-reference Wald test can therefore reject too often in skewed or low-expected-count regimes.

This article studies a deliberately small modification. Daniel's Method first applies the classical $O(n^{-1})$ bias correction to each MI estimate, studentizes their difference using empirical local-information variances, and then replaces the standard-normal reference by a Student distribution whose effective degrees of freedom use the Welch–Satterthwaite formula [4, 5]. The conceptual pipeline is shown in Fig. 1. Because every operation is computed directly from two count tables, the method is deterministic and has $O(rc)$ cost once the tables are formed.

The contribution is a synthesis and evaluation rather than the invention of its ingredients. Welch tests have previously been applied to collections of MI estimates, and asymptotic comparisons of discrete MI values already exist. The narrow contribution is, to our knowledge, the first systematic finite-sample evaluation of this specific one-table-per-population construction: bias-corrected discrete MI, empirical influence variance, and a Welch–Satterthwaite reference under the unrestricted weak null Eq. (1).

2 Related Work

2.1 Finite-sample estimation of mutual information

For an $r \times c$ discrete distribution $P = (p_{ij})$, MI is

$$\mathrm{I}(P) = \sum_{i=1}^{r} \sum_{j=1}^{c} p_{ij} \log \frac{p_{ij}}{p_{i+} p_{+j}}. \tag{2}$$

Replacing probabilities by empirical proportions gives the maximum-likelihood or plug-in estimator. Its simplicity hides a systematic finite-sample upward bias. Miller's entropy correction [6], subsequent MI-specific calculations [7], and broader analyses [8] show why sparse sampling can contaminate information estimates even after a leading correction.

For fixed positive support, the plug-in MI estimator is asymptotically normal away from independence. Its first-order variance is the variance of the local log density ratio [7]. Influence-function methods provide a general language for this expansion and for information-theoretic functionals more broadly [9]. These results motivate fast Wald inference, but the first-order influence function degenerates at exact independence, where the chi-squared rather than normal limit is relevant.

2.2 Comparing mutual information values

The idea of comparing MI across populations is not new. Mora and Ruiz-Castillo derived asymptotic inference and pairwise comparisons for a discrete MI segregation index and recommended bootstrap methods in smaller samples [10]. Hart and Giszter combined jackknife uncertainties when comparing histogram MI estimates in neuroscience [11]. These works are close mathematical predecessors to the present two-sample problem.

Other studies apply familiar mean tests to *replicated* MI estimates. Sarkar and Pandey used a Student test on MI values obtained from randomized or shuffled astronomical data [12]. Prince et al. applied Welch comparisons to per-neuron MI outcomes [13]. Martin et al. used an unequal-variance Welch test on repeated, non-overlapping partition estimates from a continuous KSG estimator [14, 15]. These studies establish that Welch testing has entered MI applications. They do not, however, derive the two variance components from one discrete multinomial table per population as done here. Mather et al. evaluated Welch and MI leakage detectors side by side, but did not apply the Welch reference to the MI estimator itself [16].

2.3 Permutation under a weak null

Raw permutation is exact when observations are exchangeable under the strong null $P = Q$. Under Eq. (1), P and Q may differ in every respect other than MI, so group labels are generally not exchangeable. Consequently, naively permuting labels can test the wrong null. General permutation theory shows that studentization can recover asymptotic validity for equality of parameters under weak assumptions while retaining finite exactness when the complete distributions are equal [17]. We therefore use a studentized table-label permutation as the resampling benchmark, rather than raw permutation or a one-table independence shuffle.

3 Daniel's Method

3.1 Data and estimand

Let $N^P = (N^P_{ij})$ and $N^Q = (N^Q_{ij})$ be independent $r \times c$ count tables with totals n_P and n_Q. Categories must be aligned between groups. Write $\widehat{p}_{ij} = N^P_{ij}/n_P$ and define $\widehat{q}_{ij}$ analogously. All formulas use natural logarithms and therefore report nats. Changing to bits multiplies the estimated difference and its standard error by the same constant, leaving the test statistic and p-value unchanged.

The parameter of interest is the signed differential MI

$$\Delta = \mathrm{I}(P) - \mathrm{I}(Q). \tag{3}$$

The test is two-sided by default, although a directional alternative follows immediately when it is scientifically pre-specified.

3.2 Bias-corrected effect estimate

Let $d = (r-1)(c-1)$. Under fixed positive support, the leading plug-in MI bias is $d/(2n)$. We therefore

Figure 1: **Overview of Daniel's Method.** Two independent contingency tables are converted to plug-in MI estimates, corrected for leading finite-sample bias, studentized with influence-function variances, and compared with a Welch-type Student reference. The target is equality of MI, not equality of the complete distributions and not independence within either table.

define

$$\widetilde{\mathrm{I}}_P = \widehat{\mathrm{I}}_P - \frac{d}{2n_P}, \tag{4}$$

$$\widetilde{\mathrm{I}}_Q = \widehat{\mathrm{I}}_Q - \frac{d}{2n_Q}, \tag{5}$$

$$\widehat{\Delta}_{\mathrm{BC}} = \widetilde{\mathrm{I}}_P - \widetilde{\mathrm{I}}_Q. \tag{6}$$

This correction is especially important when $n_P \neq n_Q$, because the uncorrected difference has leading bias $d(1/n_P - 1/n_Q)/2$ even under equal population MI.

The correction uses the declared alphabet dimensions, not the observed nonempty dimensions. It is appropriate for the stated fixed-positive-support asymptotics. In a severely sparse sample it can overcorrect, and corrected MI values may be negative. Neither clipping nor data-dependent support changes are applied because both would alter the estimand and sampling expansion.

3.3 Influence-function uncertainty

For each positive cell, define local information

$$\ell^P_{ij} = \log \frac{\widehat{p}_{ij}}{\widehat{p}_{i+}\widehat{p}_{+j}}. \tag{7}$$

The empirical influence variance is

$$\widehat{V}_P = \sum_{ij:\widehat{p}_{ij}>0} \widehat{p}_{ij} \left(\ell^P_{ij} - \widehat{\mathrm{I}}_P\right)^2, \tag{8}$$

with $\widehat{V}_Q$ defined analogously. Independence of the samples gives

$$\widehat{\mathrm{SE}} = \sqrt{\frac{\widehat{V}_P}{n_P} + \frac{\widehat{V}_Q}{n_Q}}. \tag{9}$$

The standardized statistic is

$$T = \frac{\widehat{\Delta}_{\mathrm{BC}}}{\widehat{\mathrm{SE}}}. \tag{10}$$

Equations (6)–(10) are identical to the bias-corrected normal Wald baseline.

Table 1: **Operational algorithm for one comparison.**

1	Validate two aligned, independent count tables.
2	Compute plug-in MI in each table using Eq. (2).
3	Subtract $d/(2n)$ and form the difference in Eq. (6).
4	Compute local-information variances using Eq. (8).
5	Form the standard error and statistic in Eqs. (9)–(10).
6	Compute $\widehat{\nu}$ from Eq. (12).
7	Return the Student p-value, confidence interval, and support diagnostics.

3.4 Welch–Satterthwaite reference

Set

$$a = \frac{\widehat{V}_P}{n_P}, \qquad b = \frac{\widehat{V}_Q}{n_Q}. \tag{11}$$

Daniel's Method assigns the effective degrees of freedom

$$\widehat{\nu} = \frac{(a+b)^2}{a^2/(n_P - 1) + b^2/(n_Q - 1)}. \tag{12}$$

The two-sided p-value and $100(1-\alpha)\%$ confidence interval are

$$p_{\mathrm{D}} = 2\,\Pr\{t_{\widehat{\nu}} \geq |T|\}, \tag{13}$$

$$\widehat{\Delta}_{\mathrm{BC}} \pm t_{\widehat{\nu},1-\alpha/2}\widehat{\mathrm{SE}}. \tag{14}$$

For finite $\widehat{\nu}$, Student tails are heavier than normal tails. Thus, at a fixed T, p_{D} is never smaller than the normal Wald p-value. The method is a transparent conservative adjustment, not a new effect estimator.

3.5 First-order justification and qualification

For a positive fixed table,

$$\sqrt{n}\{\widehat{\mathrm{I}} - \mathrm{I}(P)\} = \frac{1}{\sqrt{n}} \sum_{k=1}^{n} \psi_P(X_k, Y_k) + o_P(1), \quad (15)$$

where $\psi_P(x, y) = \log\{p_{xy}/(p_{x+}p_{+y})\} - \mathrm{I}(P)$ is the MI influence function. Combining the independent expansions for P and Q establishes the normal-reference Wald limit.

Proposition 1. *Assume fixed finite alphabets, positive population cell probabilities, independent and identically distributed observations within each group, independent groups, $n_P/(n_P + n_Q) \to \lambda \in (0, 1)$, and positive influence variances. Under H_0, the statistic T in Eq. (10) converges in distribution to $N(0, 1)$. Furthermore, $\widehat{\nu} \to \infty$, so the Student reference in Eq. (14) converges to the same normal limit.*

The proposition establishes first-order compatibility, not exact finite-sample validity. Classical Welch inference combines ordinary sample variances with known Gaussian degrees-of-freedom interpretations. Here, $\widehat{V}_P$ and $\widehat{V}_Q$ are nonlinear plug-in functionals: the local scores change with the same empirical tables used to estimate MI. Assigning $n_P - 1$ and $n_Q - 1$ in Eq. (12) is therefore a Welch–Satterthwaite *analogy*. Its value must be evaluated empirically.

3.6 Worked numerical example

Consider the following synthetic 3×3 tables:

$$N^P = \begin{pmatrix} 45 & 15 & 10 \\ 15 & 45 & 10 \\ 10 & 10 & 40 \end{pmatrix}, \qquad N^Q = \begin{pmatrix} 48 & 32 & 20 \\ 32 & 48 & 20 \\ 20 & 20 & 40 \end{pmatrix}.$$

Their totals are $n_P = 200$ and $n_Q = 280$, and $d = 4$. The plug-in estimates are 0.211316 and 0.054303 nats. Subtracting 0.010000 and 0.007143 gives corrected estimates 0.201316 and 0.047160, hence $\widehat{\Delta}_{\mathrm{BC}} = 0.154155$ nats.

The empirical influence variances are $\widehat{V}_P = 0.395412$ and $\widehat{V}_Q = 0.110863$. Therefore,

$$\widehat{\mathrm{SE}} = \sqrt{0.395412/200 + 0.110863/280} = 0.048713,$$

and $T = 3.16454$. Equation (12) gives $\widehat{\nu} = 278.71$. The normal p-value is 0.001553, whereas Daniel's Method returns $p = 0.001725$ and a 95% interval $[0.05826, 0.25005]$ nats. Both references reject equality, but the Student reference is slightly more conservative. The minimum independence-expected counts are 18.0 and 22.9, so this example lies comfortably inside the dense regular regime. It illustrates the calculation only and is not part of the simulation evidence.

3.7 Complexity and recommended output

Given two count tables, plug-in MI, influence variance, and support diagnostics each require a constant number of scans over rc cells. Consequently, time is $O(rc)$ and additional working memory is $O(rc)$ in the vectorized implementation. If raw observation pairs are supplied, forming the tables first adds $O(n_P + n_Q)$ time. No random number generator, resample count, or convergence tolerance is required.

A complete analysis should report n_P, n_Q, r, c, the two plug-in and bias-corrected MI estimates, $\widehat{\Delta}_{\mathrm{BC}}$, $\widehat{\mathrm{SE}}$, T, $\widehat{\nu}$, the Student p-value and confidence interval, and the normal-reference p-value as a comparator. Zero-cell fractions and expected-count diagnostics should accompany sparse applications. A noncomputable first-order variance must be reported as invalid, not silently converted to $p = 1$. The bias correction and the Welch reference are distinct: the former changes the point estimate, while the latter changes only the inferential reference.

4 Experimental Design

4.1 Weak-null population construction

Every null experiment used independent multinomial samples from population pairs satisfying $\mathrm{I}(P) = \mathrm{I}(Q)$ to numerical tolerance while allowing different margins and joint distributions. Random Dirichlet margins and interaction structures generated heterogeneous tables, and association strength was adjusted to target MI levels of 0.03, 0.07, or 0.15 nats. Common sampled table pairs were supplied to all methods, producing paired method comparisons without training, threshold fitting, or cross-method data leakage.

4.2 Decisive simulation

The main experiment contained four stages:

1. A broad grid of 144 population pairs, shapes from 2×2 to 20×20, square and rectangular tables, equal and unequal samples with ratios 1:1, 1:2, and 1:4, and 5,000 replicates per pair.
2. A targeted hard grid of 12 low-density, unequal-sample population pairs across six shapes, with 20,000 fresh replicates per pair.
3. A 26-scenario stress grid with sample sizes 20–200 and 10,000 replicates per scenario. This deliberately approached or crossed the first-order boundary.
4. Five 3×3 power scenarios with true differences 0.02, 0.05, and 0.10 nats and 10,000 replicates each.

In total, the decisive run evaluated 1,220,000 null table pairs and 50,000 power pairs. Twelve thousand hard-grid pairs also received 999 optimized studentized table permutations.

4.3 Independent adversarial holdout

After the decisive results were inspected, a separate audit fixed new population, simulation, and bootstrap seeds. It generated 72 fresh weak-null population pairs with 10,000 table pairs each, re-evaluated six frozen hard designs with 10,000 fresh table pairs each, and generated 72 strong-null $P = Q$ scenarios with 5,000 table pairs each. This holdout added 1.14 million table-pair evaluations and was designed to test the direction and generalizability of the observed effect rather than tune the method.

4.4 Comparators and metrics

The primary analytic comparator was the bias-corrected normal Wald test, which shares the estimate, standard error, and statistic with Daniel's Method. The resampling comparator was a studentized analytic table-label permutation test. An exploratory variant using $n - 1$ rather than n in the empirical variance denominator was reported separately but was not used to define the primary method.

At nominal $\alpha = 0.05$ and 0.10, we recorded false-positive rate (FPR), Wilson intervals, mean absolute FPR error (MAE), confidence-interval coverage, valid-result rate, paired rejection changes, power, and runtime. Calibration is the central criterion: at $\alpha = 0.05$, an ideal method has FPR 0.05, and smaller $|\text{FPR} - 0.05|$ is better.

5 Results

5.1 Calibration

Table 2 summarizes the decisive null experiment. In the broad regular grid, both analytic methods were well calibrated: mean FPR was 0.04977 for normal Wald and 0.04951 for Daniel's Method. MAE improved slightly from 0.00514 to 0.00504, and the proportion of scenarios within the practical 0.035–0.065 band increased from 97.22% to 97.92%.

The targeted hard grid was deliberately liberal under normal Wald. Its mean FPR fell from 0.06177 to 0.06084, and MAE fell from 0.01177 to 0.01084, a 7.9% relative reduction. Welch removed 224 normal-Wald rejections among 240,000 hard-null pairs; no Welch-only rejection is possible because the Student p-value is nested above the normal p-value. Coverage increased by 0.00093.

The stress grid shows the limit of the reference correction. Mean FPR moved toward 0.05, but scenario-level behaviour was not uniformly better. Already conservative 2×2 settings became more conservative, and 34 first-order-degenerate cases were explicitly returned as invalid. The 99.987% valid rate is high, but stress-grid FPR is conditional on a computable positive standard error.

5.2 Independent holdout

The holdout reproduced the direction of the effect without reusing the broad population pairs (Table 3). In the fresh weak-null broad grid, mean MAE gain was 0.000210 with a scenario-bootstrap 95% interval [0.000090, 0.000344]. In all six frozen hard designs, Daniel's Method improved calibration; mean gain was 0.001283 [0.000917, 0.001650]. The fresh strong-null grid also showed a small positive average gain. These results support a real conservative effect, but not uniform superiority: in the fresh broad grid, 26 scenarios improved, 14 worsened, and 32 tied.

5.3 Permutation comparison

Studentized permutation remained the most accurate comparator in the hard anchors (Table 4). Its mean FPR was 0.05458 and MAE was 0.00642, versus 0.06083 and 0.01100 for Daniel's Method on the same 12 anchor sets. Every permutation anchor was inside the practical calibration band. Thus, Daniel's Method closes only part of the gap between normal analytic inference and resampling; it does not replace permutation when maximum finite-sample calibration is worth the additional computation.

5.4 Power and computation

The heavier Student tail produces a predictable but small power cost. Across five power scenarios, mean absolute loss relative to normal Wald was 0.00154; the largest scenario-level loss was 0.0032 at $n_P = n_Q = 150$. For a 0.05-nat effect at $n = 300$, power changed from 0.2776 to 0.2761. For a 0.10-nat effect, it changed from 0.7438 to 0.7419.

Median single-pair runtime was 0.117 ms for normal Wald and 0.128 ms for Daniel's Method, a 9.1% relative overhead but only 0.011 ms in absolute terms. Across the 12 permutation anchors, median time for 999 permutations was 3.006 ms, with a range from roughly 1.1 to 5.9 ms depending on table shape. The implementation is vectorized, so the exact ratio depends on batching and hardware. The key distinction is algorithmic: the analytic calculation scans the rc cells once, whereas permutation repeats table allocation and studentized MI evaluation B times.

6 Adversarial Audit

6.1 Correctness and reproducibility

The implementation passed direct hand calculations of MI, bias, variance, degrees of freedom, statistic, and p-value. Swapping groups and relabelling rows or columns left the p-value unchanged. Student p-values were never smaller than their normal counterparts, and the two references converged at large sample size. Natural-log units were used consistently. All valid

Table 2: **Decisive null calibration at nominal $\alpha = 0.05$.** The stress grid is outside or near the intended first-order operating boundary. MAE is mean absolute false-positive-rate error.

Grid	Method	Scenarios	Mean FPR	MAE	95% coverage	In 3.5–6.5% band
Broad	Normal Wald	144	0.04977	0.00514	0.95023	97.22%
Broad	Daniel's Method	144	0.04951	0.00504	0.95049	97.92%
Hard	Normal Wald	12	0.06177	0.01177	0.93823	75.00%
Hard	Daniel's Method	12	0.06084	0.01084	0.93916	91.67%
Stress	Normal Wald	26	0.06580	0.03180	0.93420	19.23%
Stress	Daniel's Method	26	0.06276	0.03037	0.93724	15.38%

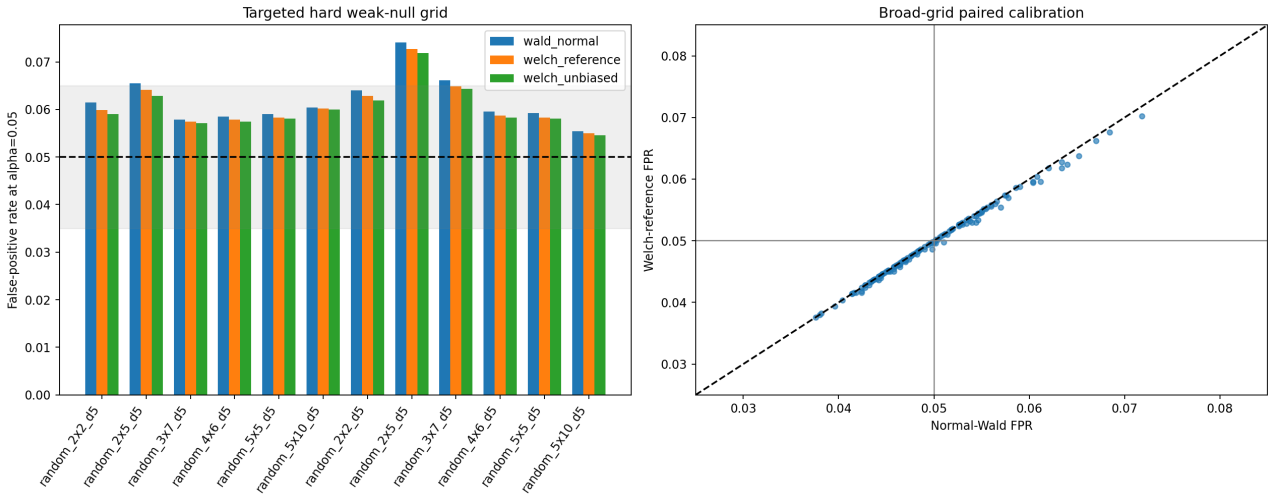

Figure 2: **Paired calibration in the decisive simulation.** Left: hard-grid FPRs for the normal baseline, primary Welch reference, and exploratory unbiased-variance sensitivity. The dashed line is nominal 0.05 and the shaded band is 0.035–0.065. Right: each point is one broad scenario; points below the diagonal indicate a lower rejection rate under Daniel's Method. The adjustment is intentionally small because hard-grid median effective degrees of freedom ranged from approximately 186 to 812.

p-values were finite and inside $[0, 1]$; malformed tables and degenerate first-order variances were rejected explicitly.

Simulation streams were separated across population generation, table sampling, permutation, and power experiments. The weak null held to a maximum numerical discrepancy of 9.3×10^{-14} nats. Reconstructing saved replicates from scenario and simulation seeds reproduced estimates and p-values to floating-point precision. No fitted hyperparameter, learned threshold, or training stage consumed validation outcomes.

6.2 Degrees-of-freedom limitation

The central theoretical qualification is visible in Fig. 4. In five representative populations, the implemented naive total degrees of freedom ranged from 171 to 858, whereas empirical moment-matched values ranged from 28 to 196. A separately derived influence function for the variance functional predicted 26 to 181 and tracked the empirical values more closely.

This discrepancy does not invalidate the first-order asymptotic result: every candidate reference converges to normal as sample size grows. It does show that Eq. (12) is not literally moment matching the sampling distribution of the nonlinear influence-variance estimator. Correlation between the estimated MI difference and its total estimated variance reached 0.84 in a skewed 2×2 audit, so improving the denominator degrees of freedom alone may still not yield an exact Student law.

6.3 Post-protocol decision

The frozen validation protocol used an internal rule requiring at least a 20% reduction in hard-grid MAE. The observed 7.9% reduction did not satisfy that rule, and the generated metadata correctly records "NO-GO." The later decision to retain the method as a thesis contribution accepts a smaller effect as scientifically useful. This is a transparent post-protocol change in materiality judgement, not a pre-specified success. The numerical result, comparators, and frozen report were not altered.

Table 3: **Adversarial holdout at nominal** $\alpha = 0.05$. Intervals are scenario-bootstrap intervals for the mean MAE gain ($\text{MAE}_N - \text{MAE}_D$).

Holdout stage	Population pairs	Tables per pair	Normal MAE	Daniel MAE	Mean gain [95% interval]
Fresh weak-null broad	72	10,000	0.004792	0.004582	0.000210 [0.000090, 0.000344]
Frozen hard design	6	10,000	0.009083	0.007800	0.001283 [0.000917, 0.001650]
Fresh strong null	72	5,000	0.005142	0.004958	0.000183 [0.000064, 0.000322]

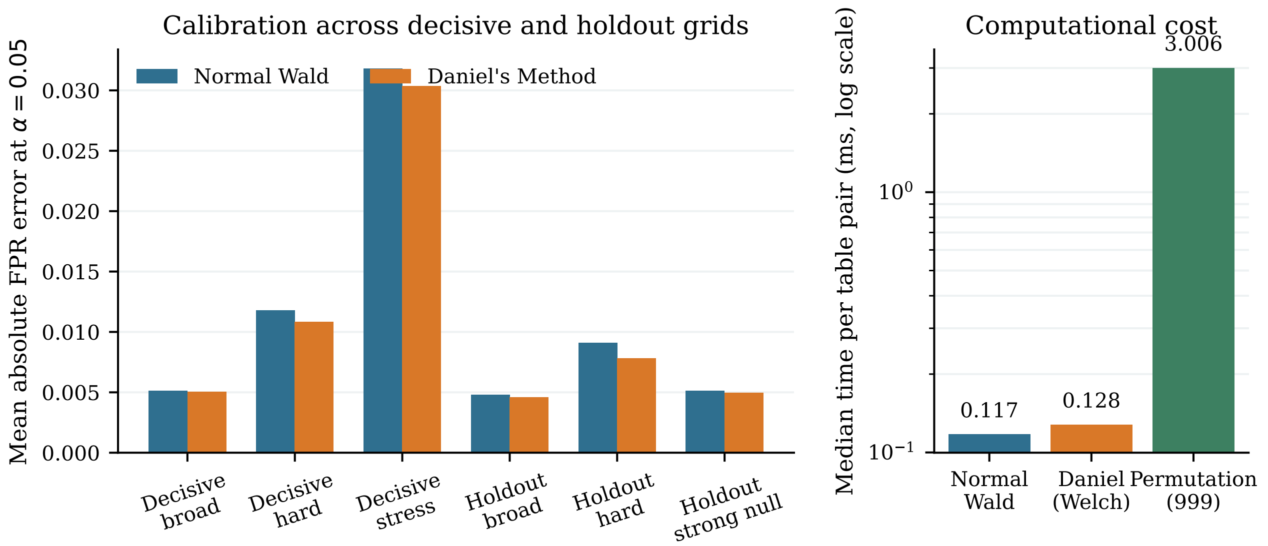

Figure 3: **Accuracy and cost summary.** Left: normal and Daniel MAE across the decisive study and independent holdout. The largest benefit occurs in deliberately liberal hard regimes; broad and strong-null effects are small. Right: median observed single-pair runtime on a logarithmic scale. The permutation result uses 999 optimized table permutations and is not JIDT's one-table independence shuffle.

Table 4: **Hard-grid permutation anchors.** Each of 12 scenarios used 1,000 table pairs and 999 studentized permutations.

Method	Mean FPR	MAE	In band
Normal Wald	0.06208	0.01208	66.67%
Daniel's Method	0.06083	0.01100	75.00%
Studentized permutation	0.05458	0.00642	100.00%

7 Discussion

7.1 What the method adds

Daniel's Method occupies a practical middle ground. Relative to normal analytic inference, it adds almost no computational burden, preserves large-sample behaviour, and provides a small conservative adjustment where the baseline is mildly liberal. Relative to studentized permutation, it is less accurately calibrated but deterministic and faster. It is most attractive in screening or repeated-comparison settings where milliseconds accumulate, exact reproducibility is valuable, and a modest calibration improvement is useful.

The size of the improvement is mathematically unsurprising. In the hard grid, median effective degrees of freedom were approximately 186–812. Student critical values at those degrees of freedom are close to normal critical values, so few decisions change. This modest effect is a feature of a backward-compatible reference correction: the estimate and uncertainty model remain untouched.

7.2 Relationship to standard MI tests

Table 5 separates four procedures that are easily conflated. Chi-squared and JIDT shuffling address independence within one population. They are appropriate for $H_0 : \mathrm{I}(P) = 0$, not Eq. (1). Normal Wald, Daniel's Method, and studentized two-sample permutation address equal MI across independent populations. Raw group-label permutation is exact only under the stronger $P = Q$ null; studentization is required for the heterogeneous weak null.

7.3 Novelty boundary

No broad claim that Welch has never been used with MI is sustainable. Published examples use Welch on repeated continuous-MI estimates [15], per-unit MI outcomes [13], and related information measures. Nor are the plugin estimator, $d/(2n)$ correction, influence variance, two-sample studentization, or Satterthwaite formula individually new.

Table 5: **How the proposed procedure relates to common significance methods.** A method is only a valid comparator when it targets the same null.

Method	Null hypothesis	Computation	Role
Chi-squared / JIDT analytic	$\mathrm{I}(P) = 0$	Closed-form reference	One-table independence; not a test of equal MI across groups.
JIDT variable shuffling	$\mathrm{I}(P) = 0$	Repeated data-label shuffles	One-table empirical independence significance; not a direct comparator.
Bias-corrected normal Wald	$\mathrm{I}(P) = \mathrm{I}(Q)$	$O(rc)$, deterministic	Historical two-sample analytic baseline.
Daniel's Method	$\mathrm{I}(P) = \mathrm{I}(Q)$	$O(rc)$, deterministic	Proposed mildly conservative finite-df refinement.
Studentized group permutation	$\mathrm{I}(P) = \mathrm{I}(Q)$ asymptotically; exact if $P = Q$	B repeated allocations and evaluations	More accurate regular resampling benchmark, with higher cost.

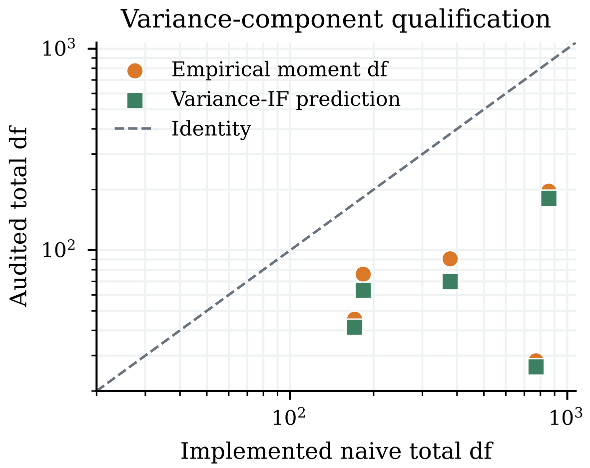

Figure 4: **The implemented degrees of freedom are heuristic.** Empirical moment degrees of freedom for the total estimated variance are well below Eq. (12). A variance-functional influence calculation tracks the audit more closely. The current procedure must therefore be described as Welch-inspired, not exact.

The defensible contribution is narrower: a complete, deterministic, MI-specific procedure for two independent discrete tables under the weak equal-MI null, together with unusually broad finite-sample calibration, permutation comparison, reproducibility checks, and an adversarial audit of the degrees-of-freedom assumption. A structured search did not locate the same combination. Before any publication-priority statement, the search should be repeated in authenticated Scopus, Web of Science, MathSciNet, zbMATH, and ProQuest databases.

7.4 Limitations and operating range

The method requires independent observations within each group and independent groups; fixed, aligned categorical alphabets; positive population support; sample-size proportions bounded away from zero and one; and nonzero population influence variance. It does not currently cover paired, clustered, longitudinal, or time-series samples.

Exact and near independence are outside scope because the first-order MI influence function degenerates. Structural zeros, unknown support, severe sampling sparsity, or alphabets growing with sample size can invalidate both the leading bias correction and variance approximation. The reported zero-cell and expected-count diagnostics describe a table but are not proven validity rules. Conditional MI, transfer entropy, and continuous MI require new derivations.

The simulation evidence is extensive in null calibration but less broad in power: the five alternatives are 3×3 tables with equal sample sizes. The permutation comparison covers 12 hard anchors with 1,000 outer replicates each. A final thesis study should broaden power across shapes and unequal samples, increase the outer permutation replicates, include $\alpha = 0.01$, and add real-data demonstrations. These additions improve external validity; simply increasing the existing null replicate counts would add little.

7.5 Next methodological step

The variance-functional audit suggests a principled extension. If τ_P^2 denotes the variance of the influence function of $\widehat{V}_P$, first-order moment matching gives a component degree of freedom of approximately $2n_P V(P)^2/\tau_P^2$, with an analogous term for Q. This prediction tracked empirical component degrees of freedom more closely than $n - 1$. It must be frozen and tested on untouched populations before replacing Eq. (12). Corrected Satterthwaite formulas proposed for general variance synthesis may also be relevant [18], but their assumptions are not yet established for MI.

8 Conclusion

Daniel's Method adapts a familiar statistical idea to a distinct information- theoretic problem: fast inference for equality of two independent discrete mutual information values. It combines leading plug-in bias correction, influence-function uncertainty, and a Welch–Satterthwaite-inspired reference without resampling. Across a large pre-specified simulation and an independent holdout, it produced a small, reproducible conservative calibration improvement over normal Wald with negligible power and runtime costs. Studentized permutation remained more accurate.

The result should be interpreted precisely. This is not an exact Welch theorem for MI, not a one-table independence test, and not a solution to severe sparsity. Its value is a clean and general regular-case refinement, an auditable implementation, and a quantitative map of where the approximation helps and where first-order MI inference remains inadequate.

Reproducibility and Data Availability

The implementation, frozen protocol, experiment runners, aggregate result tables, seeds, environment metadata, tests, and figure-generation code are stored in the accompanying research repository. The full null replicate archive is reproducible from the saved scenario definitions and seeds but is not versioned because of its size. The manuscript figures are generated by `article/make_figures.py`; the decisive experiment is generated by `experiments/run_validation.py`. The method implementation is `src/welch_differential_mi/welch.py`.

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